

Embryology

- 1) With regard to the placenta and fetal membranes, which statement is false?
- a) The finger-like projections of cytotrophoblast into syncytiotrophoblast are called primary chorionic villi
 - b) When the mesenchyme of the villi gives rise to blood vessels and blood cells, the villi is known as primary villi.
 - c) Two arteries and one vein are found in the umbilical cord.
 - d) Allantois is responsible for bladder development.
 - e) During the fourth week, the endoderm of the umbilical vesicle is incorporated into the embryo as the primordial gut.

2) The placenta has all of the following functions EXCEPT:

- a) Deoxygenation of blood
- b) Hormone production
- c) Protection
- d) Nutrition
- e) Excretion

3) At the fetomaternal junction, projections of decidua basalis towards the chorionic plate serve to divide the placenta into irregular areas called:

- a) Endometrium
- b) Cotyledons
- c) Anchoring villi
- d) Smooth chorion
- e) Villus chorion

4) During labor, what membrane ruptures ("water breaks")?

- a) Amniochorionic
- b) Amnionic
- c) Chorionic
- d) Decidua capsularis
- e) Decidua parietalis

5) Which of the following statements is not the function of the amniotic fluid?

- a) Acts as a barrier to infection.
- b) Helps control the embryo's body temperature.
- c) Assists in maintaining homeostasis of fluid and electrolytes
- d) Blood formation occurs in its wall during the third to fifth weeks.
- e) Permits normal fetal lung development

6. Most *birth defects* occur during which of the following periods/stages of the intrauterine life?

- a) First 2 weeks
- b) Embryonic period
- c) Fetal period
- d) Last trimester
- e) Zygote

The urachus forms the Median umbilical ligament

_____ 16. With regard to the placenta,

- a) the maternal part of the placenta develops from the chorionic sac.
- b) the fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
- c) amniochorionic membrane ruptures during labor.
- d) as pregnancy advances, the placental membrane becomes progressively thinner.
- e) the fetal surface of a freshly delivered placenta has the characteristic cobblestone appearance produced by cotyledons.

- _____ 17. Concerning the fetal membranes (developmental adnexa),
- a) the yolk sac helps control the embryo's body temperature.
 - b) primordial germ cells appear in the endodermal lining of the wall of the allantois.
 - c) during the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.
 - d) as the bladder enlarges, the umbilical cord involutes to form urachus.
 - e) blood formation occurs in the wall of allantois during the third to fifth weeks.

- _____ 18. Concerning multiple pregnancies,
- a) twins with one amnion, one chorion and one placenta are always monozygotic.
 - b) the rate of dizygotic twinning increases with maternal age.
 - c) the frequency of monozygotic twinning shows marked racial differences.
 - d) dizygotic twins always have two amnions and two chorions.
 - e) monozygotic twins are of the same sex.

- _____ 20. With regard to birth defects,
- a) the phenotype of Turner syndrome is female.
 - b) thalidomide induces limb defects.
 - c) trisomy of the autosomes occurs with increasing frequency as maternal age increases.
 - d) deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
 - e) phenobarbital is considered to be a safe, antiepileptic drug for use during pregnancy.

- _____ 48. Which of the following best describes the *placental components of dizygotic twins*?
- a) one placenta, two amniotic sacs, one chorion
 - b) one placenta, two amniotic sacs, two chorions
 - c) two placentas, two amniotic sacs, one chorion
 - d) two placentas, two amniotic sacs, two chorions
 - e) one placenta, two amniotic sacs, two chorions

- _____ 2. Most birth defects occur during:
- a) The first 2 weeks
 - b) The embryonic period
 - c) The fetal period
 - d) Early childhood
 - e) None of the above

- _____ 2) Most congenital malformations occur during which weeks of gestation?
- a) 0 to 3
 - b) 3 to 8
 - c) 8 to 10
 - d) 10 to 16
 - e) 16 to birth

- _____ 3) Clubfeet, which is due to mechanical forces in the womb, is an example of:
- a) Disruption
 - b) Deformation
 - c) Syndrome
 - d) Minor anomaly
 - e) Association

- _____ 4) **VACTERL (vertebral, anal, cardiac, tracheoesophageal, renal, limb) are non-random anomalies that occur together more frequently than chance alone. Which of the following is this an example of?**
- a) **Disruption**
 - b) **Deformation**
 - c) **Syndrome**
 - d) **Minor anomaly**
 - e) **Association**

- _____ 7) Which of the following statement about the principles of teratology is NOT true?

- a) Susceptibility to teratogenesis depends on the genotype of the conceptus.
- b) The maternal genome plays an important role with respect to drug metabolism, resistance to infection, and other processes that affect the conceptus.
- c) Susceptibility to teratogenesis depends on the development stage at the time of exposure.
- d) Manifestation of abnormal development depends on duration of exposure and not dose of exposure to a teratogen.
- e) Manifestations of abnormal development include death, malformation, growth retardation, and function disorders.

_____ 9) Toxoplasmosis (*Toxoplasmosis gondii*, protozoan parasite) has what characteristic feature of fetal infection?

- a) Hydrocephalus b) Cerebral calcifications c) Microphthalmia d) Limb hypoplasia e) Cataracts

_____ 10) What types of birth defects can ionizing radiation cause?

- a) Central nervous system b) Limb defects c) Gastrointestinal defects
- d) Vascular defects e) Virtually any type of defect

_____ 12) During the 6th through the 11th week of development a woman uses methamphetamine. Along with cardiovascular defects, which of the following is the most likely defect her child could be born with?

- a) Limb defects b) Anencephaly c) Cleft lip/palate
- d) Masculinization of the female genitalia e) Malformation of the uterine tubes

_____ 1. Which of the following is not the classic feature of the *Turner syndrome*?

- a. short stature b. webbed neck c. transverse palmar flexion crease (simian crease)
- d. lymphedema of the hands and feet. e. broad shieldlike chest with widely spaced nipples

_____ 2. Which of the following would cause *stained teeth* and *hypoplasia of enamel*?

- a. Aminopterin b. Tetracycline c. Thalidomide d. Penicillin
- e. Isotretinoin (13-cis-retinoic acid)

_____ 3. Which of the following does not indicate the clinical diagnosis of *achondroplasia*?

- a. short stature with bowed legs. b. short limbs and fingers but normal length of trunk.
- c. a weak catlike cry and microcephaly (abnormally small head). d. depressed nasal bridge.
- e. an increase in lumbar lordosis and a relatively large head with prominent forehead.

_____ 4. Which of the following is not a *minor anomaly*?

- a. Microtia (small ears) b. Pigmented spots c. Short palpebral fissures
- d. Curvature of the 5th finger (clinodactyly) e. None of the above

_____ 5. Which of the following does not indicate the clinical diagnosis of *Klinefelter's syndrome*?

- a. flecked, crossed-over fingers; deformed feet.
- b. small, hard testes (hypogonadism).
- c. hyalinization of seminiferous tubules and aspermatogenesis; Infertility.
- d. Sparse body hair ; gynecomastia.
- e. often tall with disproportionately long lower limbs (unproportionate growth in height).

_____ 15) Which of the following teratogens is associated with indistinct philtrum, thin upper lip, depressed nasal bridge, short nose, and flat mid-face?

- a) Syphilis
- b) Alcohol
- c) Thalidomide
- d) Lithium
- e) Hyperthermia

_____ 11) Which of the following teratogens is associated with absence of extremities?

- a) Thalidomide
- b) Rubella
- c) Syphilis
- d) Lithium
- e) Lead

_____ 13) Which of the following teratogens is the leading cause of mental retardation?

- a) Thalidomide
- b) Lithium
- c) Syphilis
- d) Alcohol
- e) Lead

_____ 14) A woman is prescribed Isotretinoin (analog of vitamin A) for her cystic acne. Although she is pregnant, the clinician does not question her about being pregnant or taking a pregnancy test. Which of the following types of birth defects can this patient's child be born with?

- a) Central nervous system
- b) Limb defects
- c) Gastrointestinal defects
- d) Vascular defects
- e) Virtually any type of defect

_____ 16) Diethylstilbestrol (DES) was once prescribed to women to prevent miscarriages. However, it has since been found to be a teratogen that causes what type of defects?

- a) Reproductive system
- b) Central nervous system
- c) Gastrointestinal
- d) Limb (partial or total absence)
- e) Virtually any type of defect

_____ 17) Which of the following defects is the most common when the mother has diabetes?

- a) Malformation of the uterus
- b) Cleft lip/palate
- c) Heart
- d) Microcephaly
- e) Cataracts/ Glaucoma

Birth Defects and Fetal Diagnosis

2) Most congenital malformations occur during which weeks of gestation?

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3) Clubfeet, which is due to mechanical forces in the womb, is an example of:

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4) *VACTERL (vertebral, anal, cardiac, tracheoesophageal, renal, limb) are non-random anomalies that occur together more frequently than chance alone. Which of the following is this an example of?*

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5) *A vascular accident leading to bowel atresia, which is a result of morphological alteration to an already formed structure, is called:*

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- b) Deformation*
- c) Syndrome*
- d) Minor anomaly*
- e) Association*

6) What is a group of anomalies occurring together that has a specific common cause?

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7) Which of the following statement about the principles of teratology is NOT true?

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8) Maternal hyperthermia, from febrile illness and even hot-tubs/saunas, is associated with all of the following birth defects EXCEPT:

- a) Anencephaly
- b) Spina bifida
- c) Mental retardation
- d) Cardiac abnormalities
- e) Deafness

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- c) Heart and neural tube defects
- d) Microcephaly
- e) Cataracts

18) Mothers with phenylketonuria (PKU) who maintain a low-phenylalanine diet prior to conception reduce their increased risk of birth defects by what percentage?

- a) 100%
- b) 80%
- c) 60%
- d) 40%
- e) 20%

19) Which of the following defects is the most likely when the mother is obese?

- a) Malformation of the uterus
- b) Cleft lip/palate
- c) Heart and neural tube defects
- d) Microcephaly
- e) Glaucoma

20) A pregnant woman is trying to maintain a healthy diet during pregnancy. For variety, she has swordfish, shark, tilefish, and pork. Weeks later the Food and Drug Administration (FDA) releases a statement showing the fish she ate had high levels of organic mercury. A consumer alert is also released for pork as the hogs were fed seedcorn, which was sprayed with a mercury-containing fungicide. Which of the following will her child most likely be born with?

- a) Pulmonary edema
- b) Spina bifida
- c) Anencephaly
- d) Mental retardation
- e) Cerebral palsy

21) Which of the following fetal therapies is usually done to the fetus directly and not to the mother?

- a) Infection treatment
- b) Fetal anemia treatment

- c) Thyroid treatment
- d) Cardiac arrhythmia treatment

22) Which of the following defects treated by fetal surgery is the most controversial because the abnormality is not life threatening?

- a) Neural tube defects
- b) Cystic (adenomatoid) lesions
- c) Congenital diaphragmatic hernia
- d) Obstructive urinary disease of the urethra